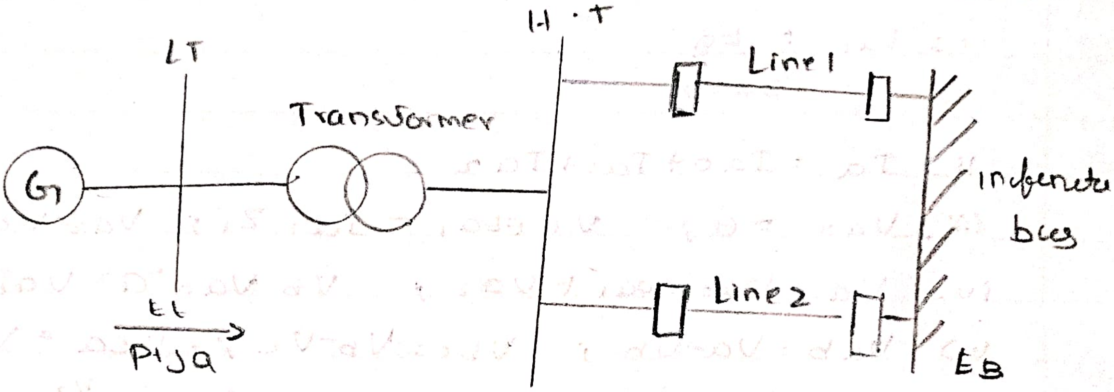
Excersise:

1. A power system comprising a thermal generating plant with four 555 MVA, 24 kV & 60 Hz units supply power to an infinite bus through a transformer and two transmission lines



the data for the system given is per unit on a base of 2220 MVA, 24 kV is given below:

$x_d' = 0.3$ pu, $H = 3.5$ MW/MVA, transformer $X = 0.15$ pu, line 1 $X = 0.5$ pu, line 2 $X = 0.93$ pu plant operating conditions $P = 0.9$ pu, p.f $= 0.9$ lag, $E_t = 1$ pu

Case 1: It is proposed to examine the transient stability of the system for a 3ϕ to ground fault at the end of the line 2 near H.T bus occurring at $t = 0$ s. the fault is cleared in 0.07 s by simultaneous opening of two CBs at the end of the both lines 2.

Case 2: Find the initial condition necessary for a classical model of the m/c for the above pre-fault operating conditions. Find the critical clearing angle and time using equal angle criterion

Aim:

to become familiar with various aspects of the transient and small signal stability analysis of single-machine infinite bus (SMIB) system,

Software required:

MATLAB 7.0

Procedure:

for a typical power system comprising a generating, step-up transformer, double circuit transmission line connected to infinite bus:

transient stability analysis:

1. Calculate the initial condition for the synchronous machine
2. Find critical clearing angle and time using equal area criterion
3. Simulate fault (apply & clear) and plot swing curve.
4. Verify critical clearing values using software (trial & error)
5. Repeat for different fault conditions and severity.
6. Determine steady state and transient stability margins

Small Signal Stability analysis:

7. Study linearized swing equation, roots damping ratio & frequencies
8. Obtain force free response for given initial condition.
9. analyze effect of positive, negative, zero damping

Given :

$X_d' = 0.3$, $X_T = 0.15$

Line1: $X_1 = 0.5$, line 2: $X_2 = 0.93$

$P_m = 0.9$, $E = 1$, $V = 1$

power factor = 0.9 lag

Step1: prefault equivalent Reactance

two lines in parallel

$$X_L = \frac{X_1 X_2}{X_1 + X_2} = \frac{0.5 \times 0.93}{0.5 + 0.93} = 0.325 \text{ pu}$$

total reactance :

$$X_{pre} = X_d' + X_T + X_L = 0.3 + 0.15 + 0.325 = 0.775$$

Step2: Initial power angle δ_0

$$P = \frac{EV}{X} \sin\delta$$

$$0.9 = \frac{1 \times 1}{0.775} \sin\delta \Rightarrow \sin\delta_0 = 0.6975$$

$$\delta_0 = 44.2°$$

Step 3: During fault

3-ϕ fault at line end -> power transfer = 0

$$P_{fault} \approx 0$$

Step 4: post-fault Reactance

one line removed -> only Line1 remains :

$$X_{post} = 0.3 + 0.15 + 0.5 = 0.95$$

maximum power :

$$P_{max} = \frac{1}{0.95} = 1.0526$$

Program:

```
% find critical angle & critical clearing time
% find frequency ; h=inertia constant ; V=terminal voltage
% E =internal voltage; dc = critical clearing angles
% tc = critical clearing time
f=60; h=5; pm = 0.8; V=1; q=0.074;
Xd = 0.31; XL = 0.21; X1=0.31;
S= complex (q, pm);
% I - current
i= S/V;
X= (0.2+0.3+(0.3/2));
a= i*x;
e = V+a;
d= abs(e);
pmax = ((d*V)/X);
delo = asin(pm/pmax);
delmax = pi-delo;
% delcc - critical clearing time angle
% tc - criticalclearingtime
delcc = acos(((pm/pmax)*(delmax-delo))+cos
        (delmax))*(180/pi)
tc = sqrt((2*h*((delcc*pi /180)-delo))/(pi*f*pm))
```

Step5 : maximum power angle δm

$$Pm = Pmax \sin \delta m$$

$$0.9 = 1.0526 \sin \delta m$$

$$\delta m = 58.7$$

Step6 : Critical clearing angle δc

using equal area criterion

$$Pm(\delta c - \delta o) = \int_{\delta c}^{\delta m} (Pmax \sin \delta - Pm)\, d\delta$$

$$0.9(\delta c - 44.2) = \int_{\delta c}^{58.7} (1.0526 \sin(44.2) - 0.9)\, d\delta$$

$$\delta c \approx 75^{\circ}$$

Step7 : critical clearing time tc

swing equation :

$$tc = \sqrt{\frac{2H}{\omega s\, pm}(\delta c - \delta o)}$$

$$\omega s = 2\pi \times 50 = 314$$

$$tc = \sqrt{\frac{2 \times 3.5}{314 \times 0.9}} \times (75 - 44.2)^{\circ}$$

$$tc \approx 0.08 \text{ sec}$$

Final answer

- initial power angle $\delta o = 44.2^{\circ}$
- maximum angle $\delta m = 58.7^{\circ}$
- critical clearing angle: $\delta c = 75^{\circ}$
- critical clearing time : $tc \approx 0.08s$

MATLAB Output:

Critical clearing angle (degrees):

89.0532

Critical Clearing Time (seconds):

0.2168

Result:

thus the various aspects of the transient and small signal stability analysis of single machine infinite bus (SMIB) system was analyzed and verified.